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AI



OpenAI launches two ‘open’ AI reasoning models

OpenAI announced Tuesday the launch of two open-weight AI reasoning models with similar capabilities [to its o-series](#). Both are freely available to download from the online developer platform [Hugging Face](#), the company said, describing the models as “state of the art” when measured across several benchmarks for comparing open models.

The models come in two sizes: a larger and more capable gpt-oss-120b model that can run on a single Nvidia GPU, and a lighter-weight gpt-oss-20b model that can run on a consumer laptop with 16GB of memory.

IMAGE CREDITS:

The launch marks OpenAI’s first ‘open’ language model since [GPT-2](#), which was released more than five years ago.

In a briefing, OpenAI said its open models will be capable of sending complex queries to [AI models in the cloud](#), as TechCrunch previously reported. That means if OpenAI’s open model is not capable of a certain task, such as processing an image, developers can connect the open model to one of the company’s more capable closed models.

While OpenAI open sourced AI models in its early days, the company has generally favored a proprietary, closed source development approach. The latter strategy has helped OpenAI build a large business selling access to its AI models via an API to enterprises and developers.

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However, CEO Sam Altman said in January he believes OpenAI has been “on the wrong side of history” when it comes to open sourcing its technologies. The company today faces growing pressure from Chinese AI labs — including DeepSeek, Alibaba’s Qwen, and Moonshot AI — which have developed several of the world’s most capable and popular open models. (While Meta previously dominated the open AI space, the company’s Llama AI models have [fallen behind](#) in the last year.)

In July, the Trump administration also urged U.S. AI developers to [open source more technology](#) to promote global adoption of AI aligned with American values.

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With the release of gpt-oss, OpenAI hopes to curry favor with developers and the Trump administration alike, both of which have watched the Chinese AI labs rise to prominence in the open source space.

“Going back to when we started in 2015, OpenAI’s mission is to ensure AGI that benefits all of humanity,” said Altman in a statement shared with TechCrunch. “To that end, we are excited for the world to be building on an open AI stack created in the United States, based on democratic values, available for free to all and for wide benefit.”

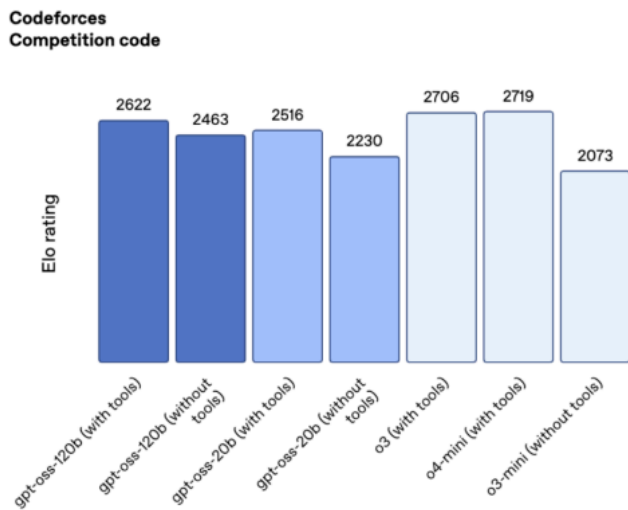


IMAGE CREDITS: TOMOHIRO OHSUMI / GETTY IMAGES

How the models performed

OpenAI aimed to make its [open model a leader](#) among other open-weight AI models, and the company claims to have done just that.

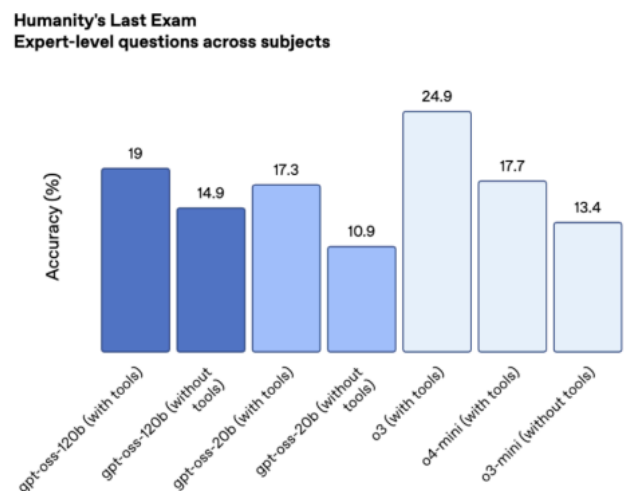
On Codeforces (with tools), a competitive coding test, gpt-oss-120b and gpt-oss-20b score 2622 and 2516, respectively, outperforming DeepSeek’s R1 while underperforming o3 and o4-mini.



OPENAI'S OPEN MODEL PERFORMANCE ON CODEFORCES.

IMAGE CREDITS:OPENAI

On Humanity's Last Exam (HLE), a challenging test of crowdsourced questions across a variety of subjects (with tools), gpt-oss-120b and gpt-oss-20b score 19% and 17.3%, respectively. Similarly, this underperforms o3 but outperforms leading open models from DeepSeek and Qwen.



OPENAI'S OPEN MODEL PERFORMANCE ON HLE.

IMAGE CREDITS:OPENAI

Notably, OpenAI's open models hallucinate significantly more than its latest AI reasoning models, o3 and o4-mini.

Hallucinations have been [getting more severe](#) in OpenAI's latest AI reasoning models, and the company previously said it doesn't quite

understand why. In a white paper, OpenAI says this is “expected, as smaller models have less world knowledge than larger frontier models and tend to hallucinate more.”

OpenAI found that gpt-oss-120b and gpt-oss-20b hallucinated in response to 49% and 53%, respectively, of questions on PersonQA, the company’s in-house benchmark for measuring the accuracy of a model’s knowledge about people. That’s more than triple the hallucination rate of OpenAI’s o1 model, which scored 16%, and higher than its o4-mini model, which scored 36%.

Training the new models

OpenAI says its open models were trained with similar processes to its proprietary models. The company says each open model leverages mixture-of-experts (MoE) to tap fewer parameters for any given question, making it run more efficiently. For gpt-oss-120b, which has 117 billion total parameters, OpenAI says the model only activates 5.1 billion parameters per token.

The company also says its open model was trained using high-compute [reinforcement learning \(RL\)](#) — a post-training process to teach AI models right from wrong in simulated environments using large clusters of Nvidia GPUs. This was also used to train OpenAI’s o-series of models, and the open models have a similar chain-of-thought process in which they take additional time and computational resources to work through their answers.

As a result of the post-training process, OpenAI says its open AI models excel at powering AI agents and are capable of calling tools such as web search or Python code execution as part of its chain-of-thought process. However, OpenAI says its open models are text-only, meaning they will not

be able to process or generate images and audio like the company's other models.

OpenAI is releasing gpt-oss-120b and gpt-oss-20b under the Apache 2.0 license, which is generally considered one of the most permissive. This license will allow enterprises to monetize OpenAI's open models without having to pay or obtain permission from the company.

However, unlike fully open source offerings from AI labs like AI2, OpenAI says it will not be releasing the training data used to create its open models. This decision is not surprising given that several active lawsuits against AI model providers, including OpenAI, have alleged that these companies inappropriately trained their AI models on copyrighted works.

OpenAI [delayed](#) the release of its open models [several](#) times in recent months, partially to address safety concerns. Beyond the company's typical safety policies, OpenAI says in a white paper that it also investigated whether bad actors could fine-tune its gpt-oss models to be more helpful in cyberattacks or the creation of biological or chemical weapons.

After testing from OpenAI and third-party evaluators, the company says gpt-oss may marginally increase biological capabilities. However, it did not find evidence that these open models could reach its "high capability" threshold for danger in these domains, even after fine-tuning.

While OpenAI's model appears to be state-of-the-art among open models, developers are eagerly awaiting the release of DeepSeek R2, its next AI reasoning model, as well as a new open model

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Microsoft brings OpenAI's smallest open model to Windows users

Ram Iyer — 5:47 AM PDT · August 6, 2025

IMAGE CREDITS:

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Microsoft is making OpenAI's new free and open GPT model, [gpt-oss-20b](#), available to Windows 11 users via [Windows AI Foundry](#), the tech giant's platform that lets users tap AI features, APIs, and popular open source models on their computers.

The company said in a [blog post](#) that gpt-oss-20b is "tool-savvy and lightweight," adding: "Optimized for agentic tasks like code execution and tool use, it runs efficiently on a range of Windows hardware, with support for more devices coming soon. It's perfect for building autonomous assistants or embedding AI into real-world workflows, even in bandwidth-constrained environments."

Launched on Tuesday, OpenAI's gpt-oss-20b can run on consumer PCs and laptops with at least 16GB of VRAM, which a modern GPU from Nvidia or Radeon would have. OpenAI said the model was trained using [high-compute reinforcement learning](#), which helps it excel at powering AI agents and call tools, such as web search or Python code execution, as part of its chain-of-thought process.

However, the model is text-only, meaning it will not be able to process or generate images and audio like the company's other models.

Also, the model has a habit of hallucinating a lot: gpt-oss-20b hallucinated in response to 53% of questions on PersonQA, OpenAI's in-house benchmark for measuring the accuracy of a model's knowledge about people.

Microsoft says it will bring the model to macOS soon, as well as other devices, though the company did not name any more. Microsoft is also making both gpt-oss-20b and older model gpt-oss-120b available via its hosted Azure AI Foundry platform.

The two new models are also [available on Amazon's AWS](#).

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Ram is a financial and tech reporter and editor. He covered North American and European M&A, equity, regulatory...

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OpenAI unveils GPT-4.5 ‘Orion,’ its largest AI model yet

Maxwell Zeff — 12:00 PM PST · February 27, 2025

Updated 2:40 pm PT: Hours after GPT-4.5’s release, OpenAI removed a line from the AI model’s white paper that said “GPT-4.5 is not a frontier AI model.” GPT-4.5’s [new white paper](#) does not include that line. You can find a link to the old white paper [here](#). The original article follows.

OpenAI announced on Thursday it is launching GPT-4.5, the much-anticipated AI model [code-named Orion](#). GPT-4.5 is OpenAI’s largest model to date, trained using more computing power and data than any of the company’s previous releases.

Despite its size, OpenAI notes in a [white paper](#) that it does not consider GPT-4.5 to be a frontier model.

Subscribers to [ChatGPT Pro](#), OpenAI’s \$200-a-month plan, will gain access to GPT-4.5 in ChatGPT starting Thursday as part of a research preview.

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Developers on paid tiers of OpenAI's API will also be able to use GPT-4.5 starting today. As for other ChatGPT users, customers signed up for [ChatGPT Plus](#) and ChatGPT Team should get the model sometime next week, an OpenAI spokesperson told TechCrunch.

The industry has held its collective breath for Orion, which some consider to be a [bellwether for the viability of traditional AI training approaches](#). GPT-4.5 was developed using the same key technique — dramatically increasing the amount of computing power and data during a “pre-training” phase called unsupervised learning — that OpenAI used to develop GPT-4, GPT-3, GPT-2, and GPT-1.

In every GPT generation before GPT-4.5, scaling up led to massive jumps in performance across domains, including mathematics, writing, and coding. Indeed, OpenAI says that GPT-4.5's increased size has given it “a deeper world knowledge” and “higher emotional intelligence.” However, there are signs that the gains from scaling up data and computing are beginning to level off. On several AI benchmarks, GPT-4.5 comes under newer AI “reasoning” models from Chinese AI startup DeepSeek, Anthropic, and OpenAI itself.

GPT-4.5 is also very expensive to run, OpenAI admits — so expensive that the company says it's evaluating whether to continue serving GPT-4.5 in its API in the long term. To access GPT-4.5's API, OpenAI is charging developers \$75 for every million input tokens (roughly 750,000 words) and \$150 for every million output tokens. Compare that to GPT-4o, which costs just \$2.50 per million input tokens and \$10 per million output tokens.

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“We’re sharing GPT-4.5 as a research preview to better understand its strengths and limitations,” said OpenAI in a blog post shared with TechCrunch. “We’re still exploring what it’s capable of and are eager to see how people use it in ways we might not have expected.”

Mixed performance

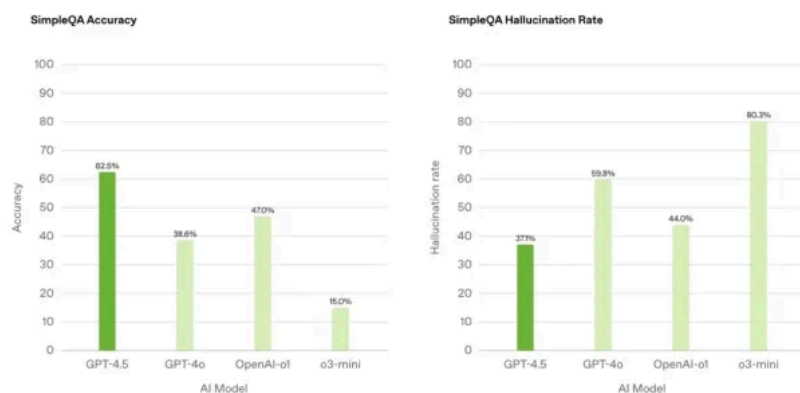
OpenAI emphasizes that GPT-4.5 is not meant to be a drop-in replacement for [GPT-4o](#), the company’s workhorse model that powers most of its API and ChatGPT. While GPT-4.5 supports features like file and image uploads and [ChatGPT’s canvas tool](#), it currently lacks capabilities like support for ChatGPT’s [realistic two-way voice mode](#).

In the plus column, GPT-4.5 is more performant than GPT-4o — and many other models besides.

On OpenAI’s SimpleQA benchmark, which tests AI models on straightforward, factual questions, GPT-4.5 outperforms GPT-4o and OpenAI’s reasoning models, [o1](#) and [o3-mini](#), in terms of accuracy. According to OpenAI, GPT-4.5 hallucinates less

frequently than most models, which in theory means it should be less likely to [make stuff up](#).

OpenAI did not list one of its top-performing AI reasoning models, deep research, on SimpleQA. An OpenAI spokesperson tells TechCrunch it has not publicly reported deep research's performance on this benchmark and claimed it's not a relevant comparison. Notably, AI startup Perplexity's Deep Research model, which performs similarly on other benchmarks to OpenAI's deep research, [outperforms GPT-4.5 on this test of factual accuracy](#).



SimpleQA measures LLM (large language model) factuality on straightforward but challenging knowledge questions.

SIMPLEQA BENCHMARKS.

IMAGE CREDITS: OPENAI

On a subset of coding problems, the SWE-Bench Verified benchmark, GPT-4.5 roughly matches the performance of GPT-4o and o3-mini but falls short of OpenAI's [deep research](#) and [Anthropic's Claude 3.7 Sonnet](#). On another coding test, OpenAI's SWE-Lancer benchmark, which measures an AI model's ability to develop full software features, GPT-4.5 outperforms both GPT-4o and o3-mini, but doesn't best deep research.

OPENAI'S SWE-BENCH VERIFIED BENCHMARK. IMAGE CREDITS:OPENAI

OPENAI'S SWE-LANCER DIAMOND BENCHMARK. IMAGE CREDITS:OPENAI

GPT-4.5 doesn't quite reach the performance of leading AI reasoning models such as o3-mini, DeepSeek's [R1](#), and [Claude 3.7 Sonnet](#) (technically a hybrid model) on difficult academic benchmarks such as AIME and GPQA. But GPT-4.5 matches or bests leading non-reasoning models on those same tests, suggesting that the model performs well on math- and science-related problems.

OpenAI also claims that GPT-4.5 is *qualitatively* superior to other models in areas that benchmarks don't capture well, like the ability to understand human intent. GPT-4.5 responds in a warmer and more natural tone, OpenAI says, and performs well on creative tasks such as writing and design.

In one informal test, OpenAI prompted GPT-4.5 and two other models, GPT-4o and o3-mini, to create a unicorn in SVG, a format for displaying graphics based on mathematical formulas and code. GPT-4.5 was the only AI model to create anything resembling a unicorn.

LEFT: GPT-4.5, MIDDLE: GPT-4O, RIGHT: O3-MINI.

IMAGE CREDITS:OPENAI

In another test, OpenAI asked GPT-4.5 and the other two models to respond to the prompt, “I’m going through a tough time after failing a test.” GPT-4o and o3-mini gave helpful information, but GPT-4.5’s response was the most socially appropriate.

“[W]e look forward to gaining a more complete picture of GPT-4.5’s capabilities through this release,” OpenAI wrote in the blog post, “because we recognize academic benchmarks don’t always reflect real-world usefulness.”

GPT-4.5’S EMOTIONAL INTELLIGENCE IN ACTION.

IMAGE CREDITS:OPENAI

Scaling laws challenged

OpenAI claims that GPT-4.5 is “at the frontier of what is possible in unsupervised learning.” That

may be true, but the model’s limitations also appear to confirm speculation from experts that pre-training “scaling laws” won’t continue to hold.

OpenAI co-founder and former chief scientist Ilya Sutskever [said in December](#) that “we’ve achieved peak data” and that “pre-training as we know it will unquestionably end.” His comments [echoed concerns](#) that AI investors, founders, and researchers [shared with TechCrunch for a feature in November](#).

In response to the pre-training hurdles, the industry — including OpenAI — has embraced reasoning models, which take longer than non-reasoning models to perform tasks but tend to be more consistent. By increasing the amount of time and computing power that AI reasoning models use to “think” through problems, AI labs are confident they can significantly improve models’ capabilities.

OpenAI plans to eventually combine its GPT series of models with its “o” reasoning series, [beginning with GPT-5 later this year](#). GPT-4.5, which [reportedly](#) was incredibly expensive to train, delayed several times, and failed to meet internal expectations, may not take the AI benchmark crown on its own. But OpenAI likely sees it as a steppingstone toward something far more powerful.

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